

Luis A. Rodriguez

Technical Architect – Embedded & Web Systems

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Summary

I'm a Senior Software Engineer with over 20 years of experience building software that ranges from videogames for small resource limited mobile devices to web applications used by millions of users. I prioritize user experiences to help them achieve whatever they're trying to do in a simpler, quicker fashion. I am capable of being an individual contributor, or leading teams where I use my organizational and emotional skills to deliver what the customer or business needs. In a mentor or lead role, I tend to be empathetic, looking for ways to keep the team's morale up so we can all be more efficient. Overall I love to design, write and deliver software that is not only technically sound, but also meaningful and user-centered.

Skills

Languages	C, JavaScript, TypeScript, CSS, HTML, Java, C++, Objective-C, Swift, z80 Assembly, Brew, J2ME
System and low level	Emscripten, Markdown, WebAssembly, Java Native Interface, Hardware Emulation, WebSerial, WebUSB, USB Communication, Cross-Compilation
Mobile/Web/Frameworks	Angular, Spring Boot, Lua, Gulp, Ant, Mocha, Jasmine, Selenium WebDriver
Leadership	Test-Driven Development, Technical Leadership, Agile/Scrum, Project Coordination, Git

Transferrable Skills

Technical Architecture	"Full-Stack Development, System Integration, Low-Level Bridging, Cross-Compilation", Architected and implemented solutions that bridge proprietary embedded systems (C/z80) with modern web applications (Angular/TypeScript) via WebAssembly and Emscripten.
Web/Front-End Expertise	"Angular, TypeScript, Responsive Design, API Integration (RESTful)", "Designed and implemented Angular applications for calculator emulation, license activation, and user utilities, utilizing third-party SSO and communicating with Java/SpringBoot middleware."
Embedded & Systems	"Hardware Emulation, Low-Level Programming (C, z80), Communication Protocols (USB, WebSerial/WebUSB, JNI)", "Designed the glue layer for targetting embedded code on the web. Established cross-platform USB communication architecture using abstraction layers and JNI."
DevOps & Automation	"Build System Optimization, CI/CD, Scripting (Gulp, Node.js, Jenkins), Deployment", "Developed and maintained complex build scripts, optimizing workflow into parallel Jenkins jobs, resulting in a 70% build time speed-up."
Technical Leadership	"Scrum Master, Technical Requirements, Cross-Functional Coordination, Code Review", "Led development teams, serving as Scrum Master and Development Lead. Authored technical requirements based on user needs. Coordinated business-wide tech sharing sessions."

Mentorship & Team Morale	"Empathetic Leadership, Mentoring, Knowledge Transfer", "Managed and mentored engineering teams, offering tutorials and code deep-dives to improve team proficiency and maintain morale."
Software Quality & Testing	"Test-Driven Development (TDD), Automation Frameworks, Code Quality Metrics", "Prototyped automation frameworks (Node.js, Mocha, Selenium WebDriver). Designed and met code quality standards (unit testing, static analyzers, mandatory code reviews)."
Product & User Focus	"User-Centered Design, Accessibility Implementation, Requirements Analysis", "Prioritized user experiences for simpler, quicker solutions. Implemented code in z80 and JavaScript to enable screen reader accessibility for calculator emulators."
Risk & Compliance	"Open Source Auditing, Security Remediation", Responsible for reviewing third-party library lists to ensure eligibility and manage licensing. Led teams to address IT security scan concerns (modifying HTTP headers).

Work Experience

Texas Instruments -- Software Architect

Tenure involved an evolution from rapid functional prototyping (reducing time-to-market) and deep systems engineering to technical leadership roles and architectural oversight. I specialized in bridging low-level hardware emulation and proprietary communication protocols with modern web applications and client product lines. Designed and met code quality metrics with the aid of tools like code reviews, unit testing and static analyzers. This career progression was formally recognized with an appointment to the first rung of TI's Technical Ladder.

Open Source Auditing

2018-2025

Responsible for reviewing software third party library lists to determine their eligibility for integrating into our products. End result of this per-release activity is a list of attributions that would get appended to the EULA for each shipped product.

Tech Sync Coordinator

2018-2025

Coordinated business-wide info sharing sessions and recruited presenters for the same. This weekly series resulted in broader understanding of deep technical topics as well as business-related subjects

Calculator Web Apps

2025

Led team that implemented the functionality for 2 web apps: One for USB connectivity with the calculator and another shell for the emulator of the calculator. Serving functions of development lead, scrum master. Reviewed user requirements and wrote technical requirements. Web apps were written in Angular, with libraries compiled from C via Emscripten. For USB connectivity, used WebSerial API.

Calculator File Transfer Library

2024-2025

Designed and implemented functionality for performing file transfers between 1) Web app and physical calculator using WebSerial API and 2) A webapp that contains an emulated calculator, making the API between both libraries match so they can be a drop-in replacement for eachother

Calculator Transformation Graphing

2024-2025

Implemented functionality to support animating graphs on new hardware platform written in C, using the APIs provided by the calculator's tools.

Calculator Build Environment

2020-2025

Developed and maintained gulp.js scripts that automated the complex process of invoking disparate platform build tools, such as CMake and MSBuild. This build system was later optimized for speed by

restructuring the workflow into parallel Jenkins jobs, resulting in a 70% build time speed-up. The task division was actively managed and rebalanced to sustain peak performance.

Calculator Device Emulation**2020-2025**

Designed and implemented the glue layer for targetting embedded code on the web by transpiling to WebAssembly. Coordinated crossfunctional team that did development and test for new features. Designed and implemented an automation entry point using WebSockets.

TI-84 Plus CE Online Calculator**2022**

Implemented Angular application that provides access to an emulator for the TI 84 Plus CE calculator on web browsers. Implemented single-sign-on usage for user authentication and license validation via 3rd party provider. This was the first publicly available TI calculator on a web app with *over 1500 users in first month.

License Activation Center**2020-2021**

Designed and implemented a key validation interface in coordination with our DBA to provide users a means to activate license codes. Front end implementation written in Angular with a RESTful middleware implemented in Java+SpringBoot communicating with an Oracle database.

TI-84 Plus CE Chrome App**2020**

In reaction to the covid pandemic, led the team that worked to support remote learning on Chrome OS allowing students coming back from spring break to have a TI calculator to work with. To date, the application has been used by over 3.5 million users. -- written in angular reusing previous work from the ExamCalc product.

Automated Test Framework Prototype**2019**

Prototyped automation framework for interacting with calculator librarires in the browser using Node.js, CucumberJS, Mocha and Selenium Web Driver.

Web Apps Server-Side**2019**

Led team in determining server configuration changes needed to address concerns brought up by IT security scans. Changes included modifying HTTP headers and additions to meta tags in the web app's HTML

Nspire Connect**2019**

Designed and implemented Angular shell to run the web app, as well as the JavaScript and C glue layers for crosscompiling NavNet, the Nspire line of calculator's communication library with Emscripten to enable communicating with an Nspire CXII device via WebUSB. 28k users since launch last year.

Examcalc Accessibiliy**2017-2018**

Implemented code needed in both z80 assembly and javascript to support market needs to make calculator emulators interact correctly with a screen reader for the vision impaired. Due to the nature of the calculator emulators, no _aria_ element is readitly available, so the emulator needs to provide the messages to be fed to a screen reader.

TI Connect For Chrome OS**2015-2016**

Ported the communications layer on a utility that talks to TI calculators using Chrome's USB API. Implemented user-facing features in AngularJS in combination with Chrome APIs.

TI Connect CE**2013-2015**

Delivered cross-platform compatibility for a proprietary CARS communication protocol by porting the codebase to generate platform-native libraries (Windows/macOS). Established the USB communication architecture using an abstraction layer and integrated the functionality into the application via Java Native Interface (JNI).

TI SmartView (Hardware Emulation)**2013-2016**

Wrote and reviewed user requirements. Designed and implemented features related to interacting with emulated hardware. Helped the development team solve memory and performance issues.

TI 84 Plus CE ExamCalc

2013

Implemented emulators for ez80-based devices which have gone through our client's pilots involving 800k users without any reported issues.

Examcalcs

2012-2013

TI calculators that are approved for use on *ACT®, SAT®, and AP® exams*, as well as most *state assessments.* . Ported and implemented calculator emulators to the web. Adapted z80 opcodes and emulated the TI-84 Silver Edition and TI-84 Plus C's ASICs. Ported Toshiba T4 cpu emulation from Java and implemented the TI-108 and TI-30MV's ASICs. Pilots involving 800k users without any reported issues.

<https://education.ti.com/en/us/about/press-center/press-listing-content-module/6-20-2013-ccss>

Document Display Standardization

2012

Given the strong desire for documents to display with the same word-wrapping and character counts on all supported devices (iPad being a new addition): researched and developed guidelines and layout rules for document rendering when being displayed on different screen resolutions.

Lua Glue

2011

Designed and implemented the infrastructure for creating new classroom activities using the Lua scripting language. The C counterpart for the Lua bindings, as well as the Lua engine itself gets cross-compiled to run on different platform including iOS, Nucleus and desktop operating systems.

CARS-SmartView

Did porting of a proprietary communications protocol that uses USB in order to have it work on a Chrome App for Chrome OS

Ui And Background Thread Refactor

Worked in conjunction with the development team to refactor the usage of JNI, improving performance and memory usage.

Image Question type

Designed and implemented some of the infrastructure necessary to create new classroom activities using the Lua scripting language. The C counterpart for the Lua bindings, as well as the Lua engine itself gets cross-compiled to run on different platform including iOS, Nucleus and desktop operating systems. The use of Lua would open up the possibility of expanding Nspire's question types in the future without needing to write them in Java

Digital Chocolate -- Game Engineering Manager

Managed the team doing porting for the US carriers' devices.

Game Engineering Manager

2009-2010

Coordinating project planning for various carriers in the US, working in conjunction with the sales and marketing teams to reduce production costs and improve revenue. Lead and mentor the engineering team, offering tutorials and code deep-dives where needed.

Brew In App Purchase Library

2010

Created client module for Qualcomm's Application Value Billing API (C/C++) in conjunction with the team in Bangalore, India. Allowing games to have in-app-purchases as well as promoting code reuse on future game titles

Star Invasion

2010

Supported team that created Alien Invaders by porting for small-screened and resource limited devices. Aided the creation team to reduce the footprint for their project; done in J2ME, targeting small-screened

and resource limited devices.

Rollercoaster Rush for iPhone

2010

Ported iPhone implementation of OpenGL game to Brew <https://itunes.apple.com/us/app/3d-rollercoaster-rush/id319575672>

LemonQuest -- iPhone Lead Programmer

Developing games and applications for the iPhone OS. Coordinated development with a geographically distributed team. Prompted the company to implement updates to infrastructure to improve productivity.

Tool Programmer

2008

Aided in the development of a stage editor for a game, done in Java.

Circulate and Circulate Prologue

2008

Originally a PC based game; it was ported to the iPhone platform. Game used every user-friendly feature on the iPhone, including it's 3d sound capabilities with Open AL and it's accelerometer. Ported the DirectX impementation to the iPhone's OpenGL ES. Translated levels originally designed for desktop/mouse to be usable on a touch device. This game earned a bronze medal from pocketgamer.co.uk.

iFit

2009

Wrote an application that provided pedometer functionality as well as fitness tips for the iPhone

Gameloft -- iPhone Programmer

Various roles, starting from Porting Engineer, Lead Developer, 3d Programmer and R&D Programmer

Blockbreaker for iPhone

2007-2008

Developed and ported first prototype for the brickbreaker game using the Celestial framework for iPhone. The Mexico studio's team was the first to build a successful prototype which led to the studio being assigned more iPhone projects

Bikini Volleyball Mini Games

2007

Implemented minigames that were included with the main volleyball game when built for devices that would support them

3d Programmer

2006-2007

In charge of porting multiple game titles for resource-limited, but 3d-capable devices. All ports used Java/J2ME

Education

Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE)

Completed coursework towards: Master's in Electronics and Telecommunications with specialization in control systems and automation, 2002-2005

Academic Project

Robot Control Over Ethernet

2004

Used RTLinux kernel modules on 2 computers to write a simple PD-controller in C with a GUI in Java on one computer to send the control signals over a TCP socket to another computer that interfaced with a 2-degree-of-freedom robot

Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM)

Bachelor's in Electronics Systems Engineering, 1997-2001

Academic Project

Control And Automation Seminar

2021

Attended summer student exchange at the Université de Poitiers, France with a focus on **control systems engineering**.

Online Courses

- iOS Persistence and Core Data - 2015
- iOS Networking with Swift - 2015
- UIKit Fundamentals I - 2015
- UIKit Fundamentals II - 2015
- Intro to iOS App Development Swift - 2015
- JavaScript Design Patterns - 2015
- Website Performance Optimization - 2014
- Object-Oriented JavaScript - 2014
- Javascript Basics - 2014
- Intro to HTML and CSS - 2014
- HTML5 Game Development - 2014
- jQuery - 2013
- JavaScript - 2013
- Software Testing - 2012
- Web Application Engineering - 2012
- Artificial Intelligence for Robotics - 2012
- Introduction to Artificial Intelligence - 2011